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0580/23

MATHEMATICS

Paper 2 Non-calculator (Extended)

October/November 2025

2 hours

You must answer on the question paper.

You will need: Geometrical instruments

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- Calculators must **not** be used in this paper.
- You may use tracing paper.
- You must show all necessary working clearly.

INFORMATION

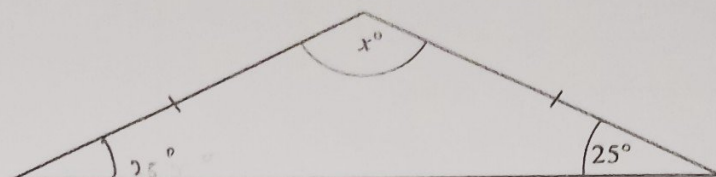
- The total mark for this paper is 100.
- The number of marks for each question or part question is shown in brackets [].

This document has **20** pages. Any blank pages are indicated.



Calculators must **not** be used in this paper.

1



NOT TO
SCALE

The diagram shows an isosceles triangle.

Find the value of x .

$$x = 180 - (50)$$

$$x = 130^\circ$$

$$x = \dots\dots\dots 130^\circ \dots\dots\dots [2]$$

- 2 Find the largest **odd** number that is a common factor of 90 and 120.

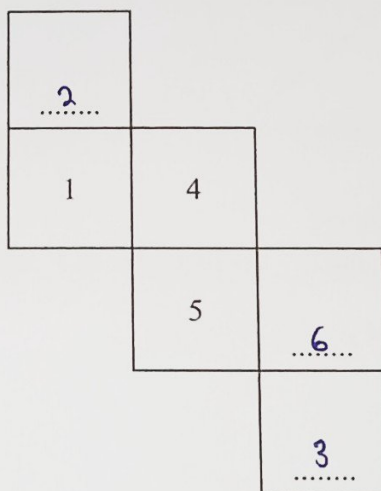
$$90 = 2 \times 3^2 \times 5 = 2 \times 9 \times 5$$

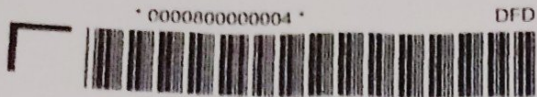
$$120 = 2^3 \times 3 \times 5 = 8 \times 3 \times 5$$

$$\dots\dots\dots \frac{1}{2} \dots\dots\dots [1]$$

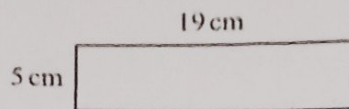
- 3 The diagram shows a net for a dice.
The dice has six faces numbered 1 to 6.
The sum of each pair of opposite faces on the dice is 7.

Write the missing numbers on the net.



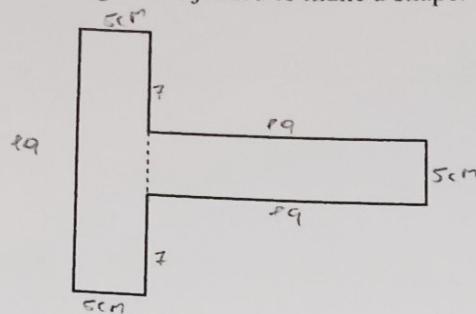


4 A rectangle measures 5 cm by 19 cm.



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Two of these rectangles are joined to make a shape.



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Work out the perimeter of the shape.

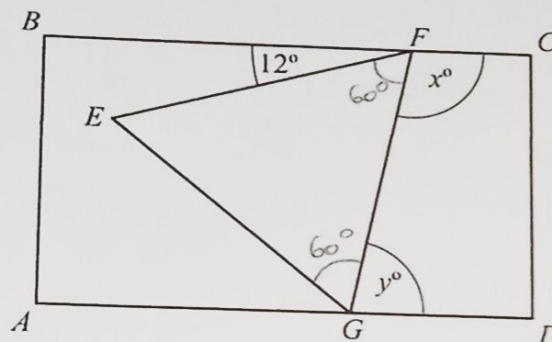
$$= (3 \times 19) + (7 \times 2) + (5 \times 9)$$

$$= 57 + 14 + 45$$

$$= 86 \text{ cm}$$

.....86..... cm [2]

5



NOT TO
SCALE

The diagram shows a rectangle $ABCD$.

EFG is an equilateral triangle that touches the rectangle at F and G .

Find the value of x and the value of y .

$$\begin{aligned} x^\circ &= 180 - (60 + 12) \\ &= 108^\circ \end{aligned}$$

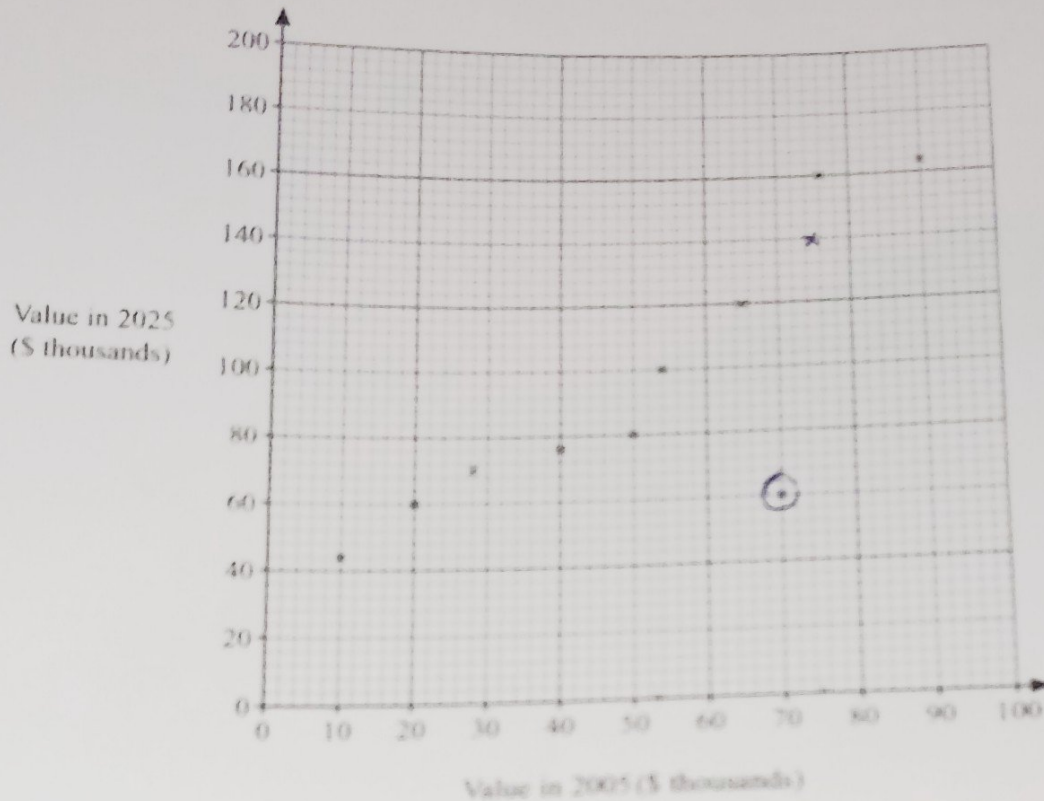
$$y^\circ = 72^\circ \text{ (co-int angles)}$$

$$x = \dots\dots\dots 108^\circ \dots\dots\dots$$

$$y = \dots\dots\dots 72^\circ \dots\dots\dots$$

[2]

- 6 The scatter diagram shows the value, in thousands of dollars, of ten paintings in 2005 and the value of the same paintings in 2025.



- (a) The value of one of the paintings in 2025 is less than expected.

Draw a circle around the point that represents this painting.

[1]

- (b) Another painting had a value of \$75 000 in 2005 and \$140 000 in 2025.

On the scatter diagram, plot this point.

[1]

- (c) Write down the number of paintings with a value of less than \$55 000 in 2005.

5

[1]

- (d) What type of correlation is shown on the scatter diagram?

Positive correlation [1]



- Find the sum of all the prime numbers less than 16.

$$26 - 2, 3, 5, 7, 11, 13$$

48 [2]

- (b) x is an integer.

The sum of the prime numbers greater than 6 and less than x is equal to 18.

Find a possible value for x .

$$x = \dots\dots\dots 12 \dots\dots\dots [1]$$

8 $\overrightarrow{DE} = \begin{pmatrix} 3 \\ -4 \end{pmatrix}$

- (a) Find $5\overrightarrow{DE}$.

$$\begin{pmatrix} 13 \\ -20 \end{pmatrix} [1]$$

- (b) Find $|\overrightarrow{DE}|$.

$$3^2 + (-4^2) = c^2 \quad b = c$$

$$9 + 16 = c^2$$

$$\sqrt{25} = c$$

..... 5 [2]

- (c) D is the point $(-2, -3)$.

Find the coordinates of the point E .

(9 , -7) [2]

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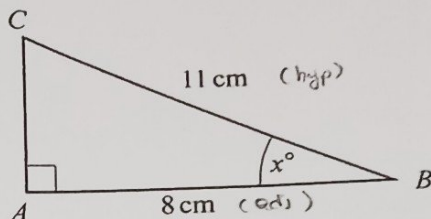
9

$$n^{15} \div n^x = n^5$$

Find the value of x .

$$x = \dots\dots\dots 10 \dots\dots\dots [1]$$

10



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	0	30	45	60	90
sin	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
cos	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0

The diagram shows a right-angled triangle ABC .

(a) Work out the exact length of AC .

$$\begin{aligned} \cos \theta &= \frac{\text{Adj}}{\text{hyp}} & a^2 + b^2 &= c^2 \\ \theta &= & 8^2 + b^2 &= 11^2 \\ & & b^2 &= 11^2 - 8^2 \\ & & b &= \sqrt{121 - 64} \end{aligned}$$

$$\dots\dots\dots \sqrt{56} \dots\dots\dots \text{cm} [3]$$

(b) $\cos x = k$

Write down the value of k .

$$k = \dots\dots\dots \frac{8}{11} \dots\dots\dots [1]$$



- 11 Sarah rolls a fair 6-sided dice twice.

Find the probability she rolls a number greater than 4 both times.

$$\begin{aligned} &= \frac{1}{6} \times \frac{1}{6} = \frac{1}{3} \times \frac{1}{3} \\ &= \frac{1}{36} = \frac{1}{9} \end{aligned}$$

$$\frac{1}{9} \dots\dots\dots [2]$$

- 12 (a) Write down the value of 73^0 .

$$1 \dots\dots\dots [1]$$

- (b) Find the value of $\frac{7}{3^{-2}}$.

$$\begin{aligned} &= \frac{7}{\frac{1}{9}} = 7 \div \frac{1}{9} = 7 \times 9 \\ &= 63 \end{aligned}$$

$$63 \dots\dots\dots [2]$$

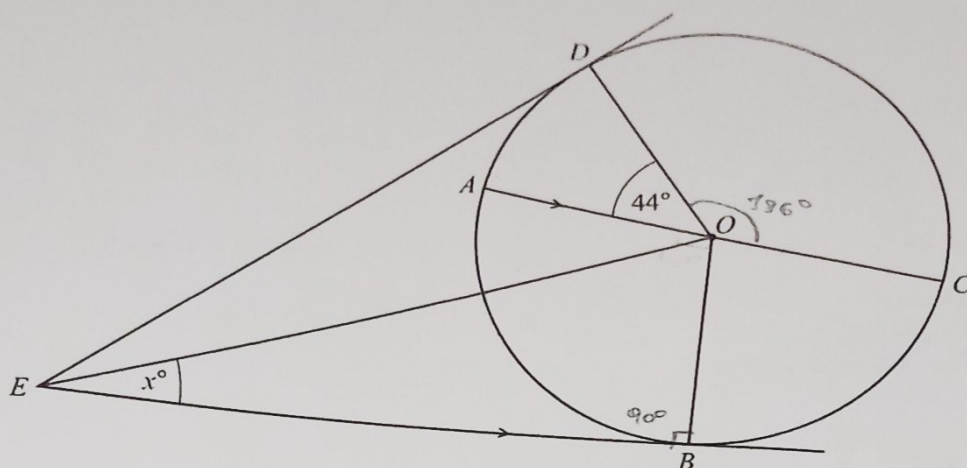
- (c) Write 27×81^2 in the form 3^n .

$$= 3^3 \times (3^4)^2$$

$$= 3^3 \times 3^8$$

$$= 3^9$$

$$3^9 \dots\dots\dots [2]$$



A, B, C and D are points on the circumference of a circle with centre O .
 ED and EB are tangents to the circle.
 AC is parallel to EB .
 Angle $AOD = 44^\circ$.

Find the value of x .

$x = \dots\dots\dots$ [4]

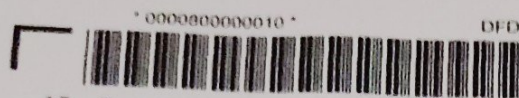
14 The minimum point on a quadratic curve is $(-3, -5)$.

(a) Find the equation of the line of symmetry of the curve.

$\dots\dots\dots x = -3 \dots\dots\dots$ [1]

(b) Write the equation of the curve in the form $y = (x + a)^2 + b$.

$y = \dots\dots\dots (x + 3)^2 - 5 \dots\dots\dots$ [1]



15 Factorise.

(a) $x^2 - 7x + 12$

$$= x^2 - 4x - 3x + 12$$

$$= x(x-4) - 3(x-4)$$

$$= (x-3)(x-4)$$

..... $(x-3)(x-4)$ [2]

(b) $5x + 10y + 6ny + 3nx$

$$= 6ny + 10y + 8nx + 5x$$

$$= 2y(3n+5) + x(8n+5)$$

$$= (2y+x)(3n+5)$$

..... $(2y+x)(3n+5)$ [2]

16 The table shows three sequences.

	1st term	2nd term	3rd term	4th term	5th term	n th term
Sequence A	8	13	18	23	28	$5n + 3$
Sequence B	$\frac{3}{2}$	$\frac{4}{3}$	$\frac{5}{4}$	$\frac{6}{5}$	$\frac{7}{6}$	$\frac{n+2}{n+1}$
Sequence C	$\frac{1}{4}$	$\frac{1}{2}$	1	2	4	$\frac{1}{4} \times 2^{n-1}$

Complete the table to show the n th term of each sequence.

A

$$= 5n + 3$$

$$= 8, 13, 18, 23, 28$$

$\uparrow$ $\uparrow$ $\uparrow$ $\uparrow$
 $+5$ $+5$ $+5$ $+5$

$$= 5n + 3$$

B

$$= \frac{3}{2}, \frac{4}{3}, \frac{5}{4}, \frac{6}{5}, \frac{7}{6}$$

$\uparrow$ $\uparrow$ $\uparrow$ $\uparrow$
 $+1$ $+1$ $+1$ $+1$

$$= \frac{n+2}{n+1}$$

C

$$= \frac{1}{4} \times 2^{n-1}$$

$$= \frac{1}{4} \times 2^{n-1}$$

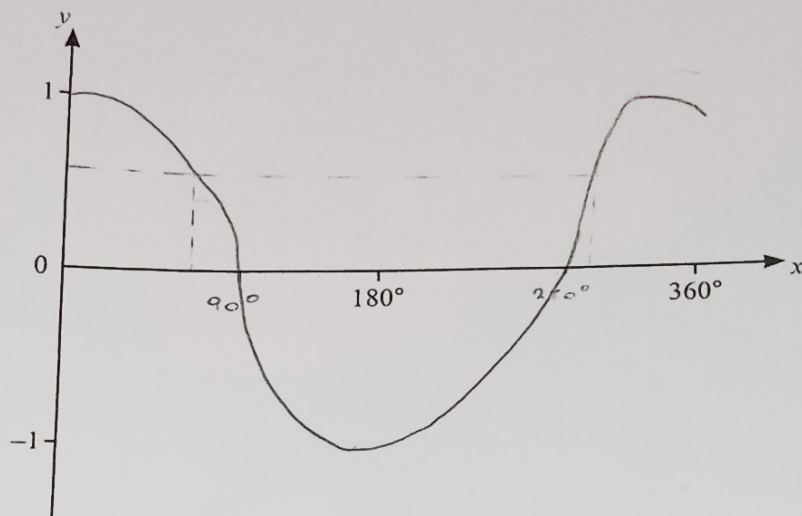
[5]

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11

- 17 (a) Sketch the graph of $y = \cos x$ for $0^\circ \leq x \leq 360^\circ$.



[2]

- (b) $\cos x^\circ = \frac{1}{\sqrt{2}}$ and x is a **reflex** angle.

Find the value of x .

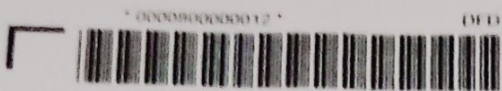
$$x = \cos^{-1} \left(\frac{1}{\sqrt{2}} \right)$$

$$x = 45^\circ$$

$$x = 270 + 45$$

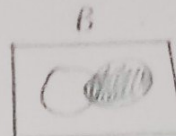
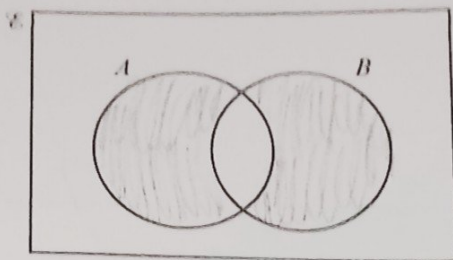
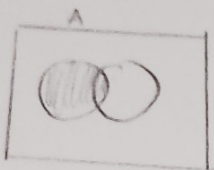
$$= 315^\circ$$

$$x = 315^\circ \quad [2]$$

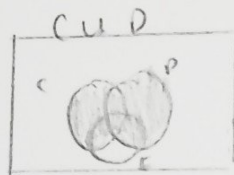
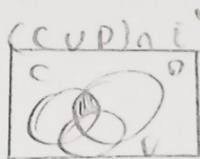
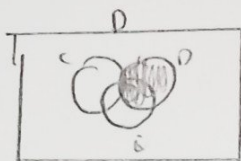
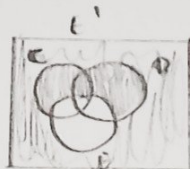
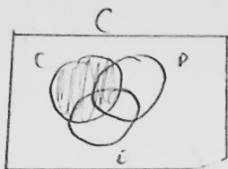
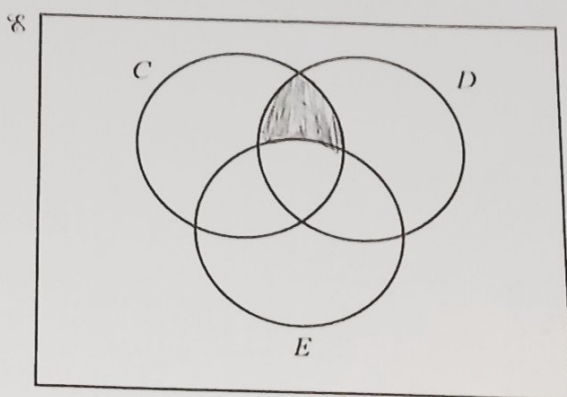


18 Shade the region in each Venn diagram.

(a) $(A \cap B)^c$



(b) $(C \cup D) \cap E^c$

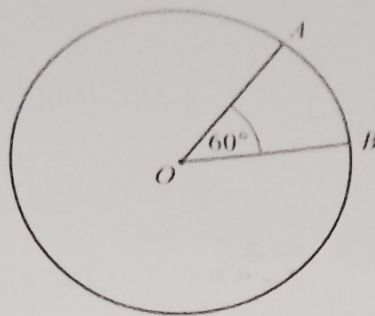


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The diagram shows a circle, centre O , radius 12 cm.

- (a) Work out the area of the **minor** sector AOB .
Give your answer in terms of π in its simplest form.

$$= \frac{60}{360} \times \pi r^2 = \frac{60}{360} \times \pi 12^2 = 24\pi$$

$$= \frac{60}{360} \times \pi 12^2 = \frac{244\pi}{6}$$

..... 24π cm^2 [2]

- (b) Calculate the length of the **major** arc AB .
Give your answer in terms of π in its simplest form.

$$= \frac{300}{360} \times 2\pi r$$

$$= \frac{300}{360} \times 2\pi 12$$

$$= \frac{5}{3} \times 24\pi$$

..... 20π cm [3]

$$= 20\pi$$

20 Work out $0.114 \div 0.2$.
Give your answer as a fraction.

$$\times 1000 \left(\begin{array}{l} r = 0.114 \\ \text{cor} = 114.54 \end{array} \right) \times 1000$$

$$r = \frac{114}{990}$$

$$= \frac{114}{990} + \frac{0.2 \times 990}{1 \times 990}$$

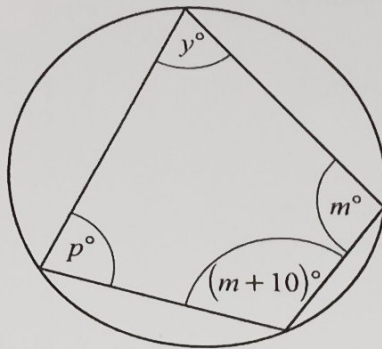
$$= \frac{114 + 198}{990}$$

$$= \frac{312}{990}$$

$$\frac{312}{990}$$

[3]

21



NOT TO
SCALE

The diagram shows a cyclic quadrilateral.
The ratio $p : m = 2 : 3$.

Find the value of y .

p and m

$$2 + 3 = 5$$

$$180 \div 5 = 36^\circ$$

$$p^\circ = 36 \times 2$$

$$= 72^\circ$$

$$m^\circ = 36 \times 3$$

$$= 108^\circ$$

y

$$(108 + 180) + y = 180$$

$$288 + y = 180$$

$$y = 180 - 288$$

$$y = 62^\circ$$

$$y = \dots\dots\dots 62^\circ \dots\dots\dots$$

- 22 The equation of line L is $y = -\frac{1}{2}x + 7$.

Find an equation of the line perpendicular to line L that passes through the point $(3, 5)$.
Give your answer in the form $y = mx + c$.

gradient

$$-\frac{1}{2} \times m_2 = -1$$

$$m_2 = -1 \div \frac{-1}{2}$$

$$\underline{m_2 = 2}$$

y-intercept

$$5 = 2(3) + c$$

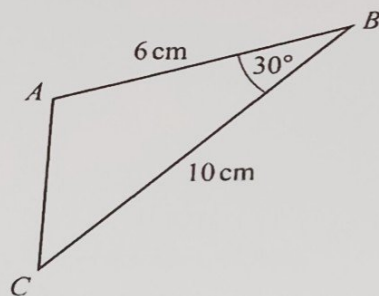
$$5 = 6 + c$$

$$5 - 6 = c$$

$$\underline{-1 = c}$$

$$y = \underline{2x - 1} \quad [3]$$

23



NOT TO
SCALE

Work out the area of triangle ABC .

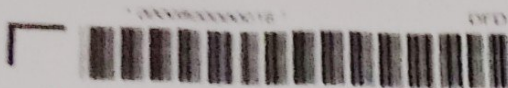
$$= \frac{1}{2} ab \sin c$$

$$= \frac{10 \times 6}{2} \sin (30)$$

$$= 30 \times \frac{1}{2}$$

$$= 15 \text{ cm}^2$$

$$\underline{15} \text{ cm}^2 \quad [3]$$



24 (a) Simplify.

$$\sqrt{125} - \sqrt{20}$$

$$\sqrt{5} \times \sqrt{25} - \sqrt{4} \times \sqrt{5}$$

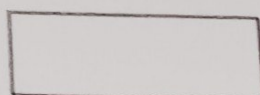
$$5\sqrt{5} - 2\sqrt{5}$$

$$3\sqrt{5}$$

$$3\sqrt{5}$$

[2]

(b)



$$2\sqrt{2} \text{ cm}$$

NOT TO
SCALE

$$3\sqrt{8} \text{ cm}$$

The area of this rectangle is $k \text{ cm}^2$.Work out the value of k .

$$= 3\sqrt{8} \times 2\sqrt{2}$$

$$= 6\sqrt{16}$$

$$= 6$$

$$k = 6\sqrt{16} \quad [2]$$

(c) Rationalise the denominator.

$$\frac{1}{\sqrt{7}+2}$$

$$= \frac{1}{\sqrt{7}+2} \times \frac{\sqrt{7}-2}{\sqrt{7}-2}$$

$$= \frac{\sqrt{7}-2}{\sqrt{7}(\sqrt{7}-2) + 2(\sqrt{7}-2)}$$

$$= \frac{\sqrt{7}-2}{7-2\sqrt{7}+2\sqrt{7}-4}$$

$$= \frac{\sqrt{7}-2}{3}$$

$$\frac{\sqrt{7}-2}{3}$$

- 25 Mahir picks one number at random from the numbers 5, 10 and 15. He then picks one number at random from the numbers 4, 5 and 6. He adds the two numbers.

The sample space diagram shows some of the possible outcomes.

		First number		
		5	10	15
Second number	4	9	14	19
	5	10	15	20
	6	11	16	21

- (a) Complete the sample space diagram. [1]
- (b) Given that the total of the two numbers is odd, find the probability that one of the numbers added is 15.

$$\frac{2}{5}$$

..... [2]

- 26 Write as a single fraction in its simplest form.

$$(a) \frac{mp}{25y} \times \frac{15}{m} = \frac{3}{5} \frac{p}{y}$$

$$\frac{3p}{5y}$$

..... [2]

$$(b) \frac{3}{2x-5} + \frac{4}{x-3}$$

$$= \frac{3(x-3)}{(2x-5)(x-3)} + \frac{4(2x-5)}{(2x-5)(x-3)}$$

$$= \frac{3x - 9 + 8x - 20}{(2x-5)(x-3)}$$

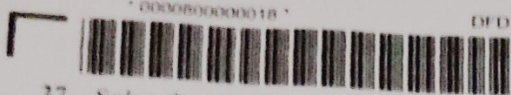
$$= \frac{11x - 29}{2x^2 - 11x + 15}$$

$$2x(x-3) - 5(x-3)$$

$$2x^2 - 6x - 5x + 15$$

$$\frac{11x - 29}{2x^2 - 11x + 15}$$

..... [3]



27 Solve the simultaneous equations.

$$y = x^2 - 8x + 22$$

$$y + 2 = 3x$$

$$y = 3x - 2$$

$$3x - 2 = x^2 - 8x + 22$$

$$0 = x^2 - 8x - 3x + 22 + 2$$

$$0 = x^2 - 11x + 24$$

$$x^2 - 8x - 3x + 24$$

$$x(x - 8) - 3(x - 8)$$

$$(x - 8)(x - 3)$$

$$\underline{x = 8} \quad \text{or} \quad \underline{x = 3}$$

$$\underline{\text{when } x = 8}$$

$$y = 8(8) - 2$$

$$= \underline{22}$$

$$\underline{\text{when } x = 3}$$

$$y = 8(3) - 2$$

$$= \underline{7}$$

$$x = \underline{8}, y = \underline{22}$$

$$x = \underline{3}, y = \underline{7}$$

[6]

28 Simplify.

$$\frac{2x^2 - 11x - 21}{x^2 - 49}$$

$$= \frac{2x^2 - 14x + 8x - 21}{(x + 7)(x - 7)}$$

$$= \frac{2x(x - 7) + 8(x - 7)}{(x + 7)(x - 7)}$$

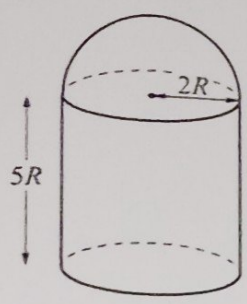
$$= \frac{(2x + 8)(\cancel{x - 7})}{(x + 7)(\cancel{x - 7})} = \frac{2x + 8}{x + 7}$$

$$\frac{2x + 8}{x + 7}$$

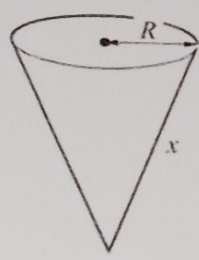
$$\frac{2x + 8}{x + 7}$$

[4]

29 In this question all measurements are in centimetres.



Solid A



Solid B

NOT TO SCALE

Solid A is made from a cylinder and a hemisphere, both of radius $2R$.
The cylinder has height $5R$.
Solid B is a cone of radius R and sloping edge x .

The **total** surface area of solid A is equal to the **total** surface area of solid B.

Find R in terms of x .

Solid A (TSA)

$$= 2\pi rh + \pi r^2 + \frac{4\pi r^2}{2}$$

$$= 2\pi 2R \times 5R + \pi 2R^2 + \frac{4\pi 2R^2}{2}$$

$$= 2\pi 10R^2 + \pi 4R^2 + 4\pi 2R^2$$

$R = \frac{x}{\sqrt{5}} \quad [5]$